

Yong-chan Park

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About Me

I am a Ph.D. student majoring in Computer Science and Engineering at [Seoul National University](#) (SNU), advised by [Prof. U Kang](#) in [Data Mining Laboratory](#). My research interests include machine learning, tensor analysis, and anomaly detection.

Research Interests

Machine learning, Tensor analysis, Anomaly detection

Education

Ph.D. Student Mar. 2020 - Present

- Computer Science and Engineering, SNU

Bachelor of Science Mar. 2012 - Feb. 2019

- Department of Mathematical Sciences, SNU

Research Experiences

A novel approach to Fourier Transform [1] June 2020 - Present

- Propose Partial Fourier Transform (PFT) that rapidly computes a part of Fourier coefficients of a given time series and demonstrate the accuracy and efficacy of PFT on real-world anomaly detection [1]. PFT achieves 20x faster running time compared to SOTA algorithms.
- Patent pending

Stock movement prediction [2] Sep. 2020 - Present

- Propose Data-Axis Transformer with Multi-Level Contexts (DTML) for stock movement prediction that learns the correlations between stocks in an end-to-end way [2]. DTML achieves SOTA accuracy on six datasets collected from US, China, Japan, and UK.

Software development for patent management Sep. 2020 - Present

- Implement LSH (Locality Sensitive Hashing) algorithm for finding similar patent data, and a BERT-based model for automatic patent classification.
- Develop a GUI with additional functionality of visualizing results.
- Industry-Academy Cooperation with Daewoo Shipbuilding & Marine Engineering Co., Ltd.

Publications

- [1] **Yong-chan Park**, Jun-Gi Jang, and U Kang. “Fast and Accurate Partial Fourier Transform for Time Series Data.” KDD 2021
- [2] Jaemin Yoo, Yejun Soun, **Yong-chan Park**, and U Kang. “Accurate Multivariate Stock Movement Prediction via Data-Axis Transformer with Multi-Level Contexts.” KDD 2021
- [3] **Yong-chan Park***, Sangjun Son*, Minyong Cho, and U Kang. “DAO-CP: Data-Adaptive Online CP Decomposition for Tensor Stream.” PLOS One 2022 (*equal contribution)
- [4] Jun-Gi Jang, Jeongyoung Lee, **Yong-chan Park**, and U Kang. “Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors - Algorithm and Application.” KDD 2023
- [5] **Yong-chan Park**, Jongjin Kim, and U Kang. “Fast Multidimensional Partial Fourier Transform with Automatic Hyperparameter Selection.” KDD 2024
- [6] Jun-Gi Jang, **Yong-chan Park**, and U Kang. “Fast and Accurate PARAFAC2 Decomposition for Time Range Queries on Irregular Tensors.” CIKM 2024
- [7] JinGee Kim, **Yong-chan Park**, Jaemin Hong, and U Kang. “Accurate Stock Movement Prediction via Multi-Scale and Multi-Domain Modeling.” BigData 2024
- [8] SeungJoo Lee, **Yong-chan Park**, and U Kang. “Accurate Coupled Tensor Factorization with Knowledge Graph.” BigData 2024

Under Review

- [R1] **Yong-chan Park**, SeungJoo Lee, and U Kang. “Fast and Accurate Online Coupled Matrix-Tensor Factorization via Frequency Regularization.”
- [R2] **Yong-chan Park**, Kisoo Kim, and U Kang. “PuzzleTensor: A Method-Agnostic Data Transformation for Compact Tensor Factorization.”